FAKE NEWS DETECTION USING MACHINE LEARNING

A project report using python and scikit-learn

ABSTRACT

Fake news has become a major challenge in the digital era, influencing public opinion and spreading misinformation rapidly. This project aims to build a machine learning model to automatically classify news articles as 'True' or 'Fake'. Using a dataset of 44,898 news articles, we implemented a PassiveAggressiveClassifier with TF-IDF Vectorization. The model achieved an exceptional accuracy of 99.55% on test data, correctly predicting 8940 out of 8980 articles. This system demonstrates high reliability and can be further developed into a real-time news verification tool to combat misinformation.

Technology: python, scikit-learn, pandas, NLP

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Objective:

To develop a machine learning model that classifies news articles as 'True' or 'Fake' using Natural Language Processing techniques

Dataset:

- Source: Kaggle Fake News Dataset
- Files: True.csv - 21,417 articles, Fake.csv - 23,481 articles
- Total Size: 44,898 news articles
- Columns Used: text for prediction, label added for classification

Methodology:

- Data Preprocessing: Removed missing values, combined both datasets
- Labeling: True News = 1, Fake News = 0
- Train-Test Split: 80% Training - 35,918, 20% Testing - 8,980
- Vectorization: TF-IDF Vectorizer with English stopwords removed, max_df=0.7
- Model: Passive Aggressive Classifier with max_iter=50
- Evaluation: Accuracy Score & Confusion Matrix

Results:

Accuracy Achieved: 99.55%

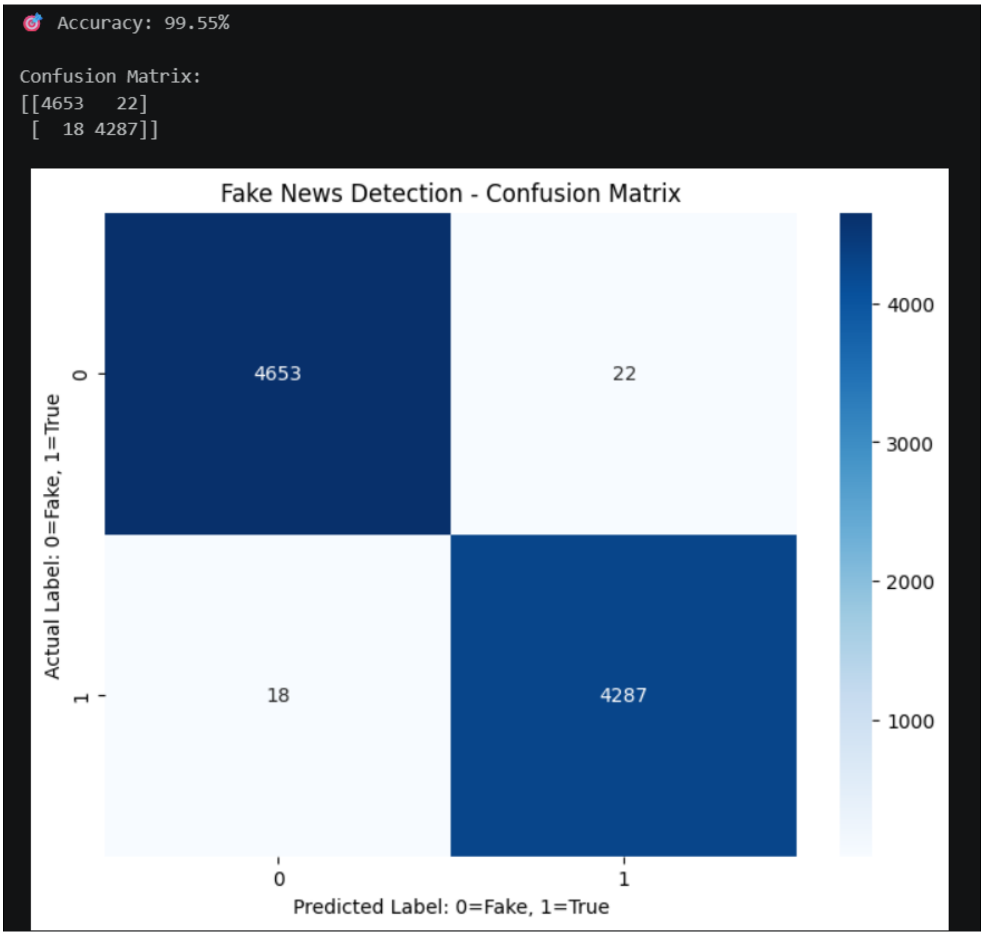
Correct Predictions: 8940 out of 8980 test samples

Confusion Matrix:

[[4653   22]

 [18 4287]]

- True Negatives - Fake news correctly identified: 4653
- False Positives - Fake predicted as True: 22
- False Negatives - True predicted as Fake: 18
- True Positives - True news correctly identified: 4287



Technologies Used:

Python 3.14, Pandas, Numpy, Scikit-learn, NLTK, Matplotlib, Seaborn, Jupyter Notebook

Conclusion:

The PassiveAggressiveClassifier model performed exceptionally well with 99.55% accuracy. This shows that TF-IDF + PassiveAggressive is highly effective for fake news detection. The model demonstrates strong capability to distinguish between real and fake news. This system can be deployed as a web application for real-time news verification.